IBM Developer
SKILLS NETWORK

SpaceX Falcon 9 Landing Success Analysis

Winning Space Race with Data Science

Public-data reproduction | Historical launch snapshot

Reproducible analysis draft
Public snapshots; no submitter identity supplied

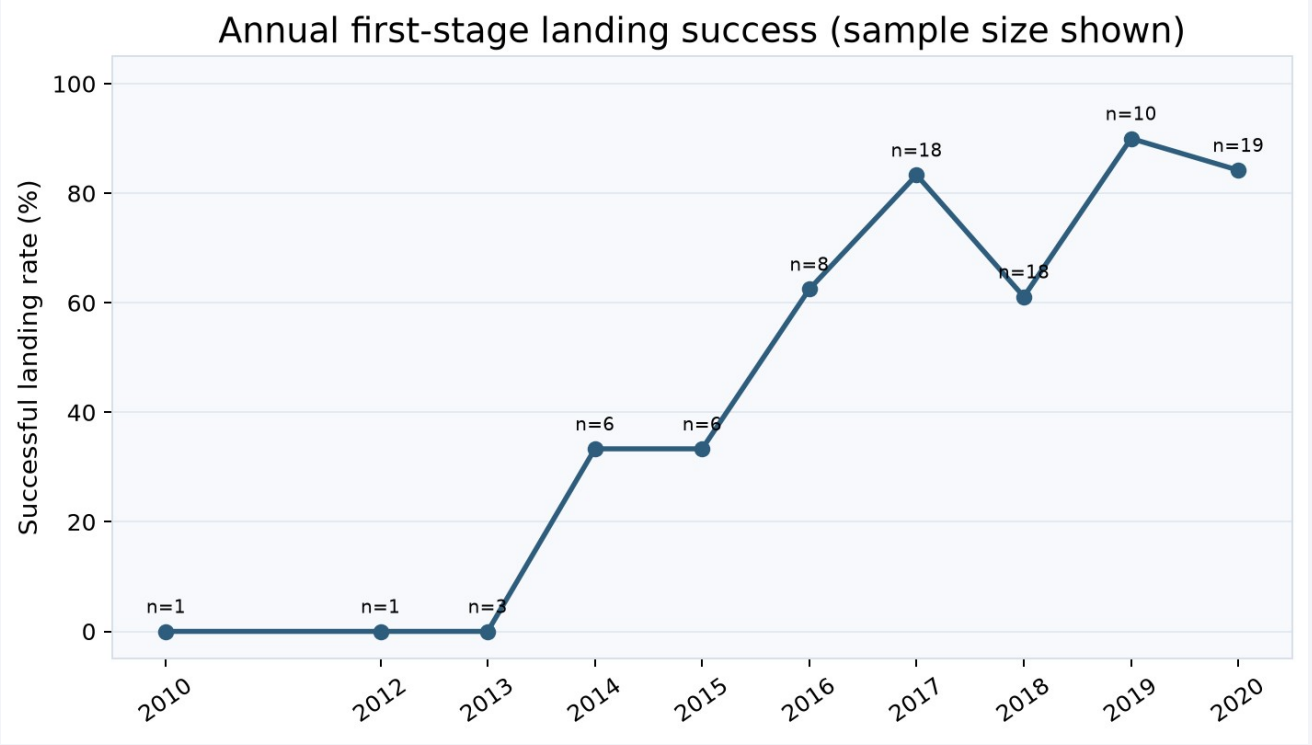


Outline

- Executive summary
- Introduction
- Methodology
- EDA and SQL results
- Folium maps
- Interactive dashboard
- Classification
- Conclusion and sources

Executive Summary

- Question: Can launch history help explain or predict first-stage landing success?
- Observed success: 60 of 90 launches (66.7%).
- Site rates: SLC 40 60.0%; KSC 77.3%; VAFB 76.9%.
- Insight: SLC 40 accounts for 55% of all successful recoveries by volume, yet has the lowest within-site rate; volume share and success rate tell different stories.
- Best CV: Random Forest 76.2% mean accuracy; held-out accuracy 88.9% (18 cases).
- Small historical sample; associations are descriptive, not causal.



Introduction

- Reusable first stages can reduce launch cost, but recovery attempts have varied over time and by mission.
- This study examines Falcon 9 first-stage landing outcomes using public historical snapshots.
- Questions: How do outcomes vary by site, orbit, payload, and launch sequence? What can a small-data classifier predict?
- Scope: API-derived launch records from 2010-2020; a separate 2010-2021 web table supports SQL exercises.



01 | METHODOLOGY

Methodology

Methodology

1. Two fixed public CSV snapshots: 90 API-derived Falcon 9 launches and 121 web-table launch rows.

- 2. Clean dates, payload units, missing values, and outcome labels; join by date (89 of 90 match).
- 3. Explore site, orbit, payload, and year patterns with charts; query the web table in SQLite.
- 4. Map launch locations and build an interactive Plotly Dash view.
- 5. Compare five classification pipelines with training-only repeated stratified CV and a held-out test split.
- No personal notebook or submission link was supplied.

Data Collection

API-derived CSV: 90 Falcon 9 launches dated 2010-06-04 to 2020-11-05; includes payload, orbit, site, booster-use, and landing outcome.

Web-table CSV: 121 launch rows dated 2010-06-04 to 2021-06-06; includes customer, booster version, landing outcome, and payload mass.

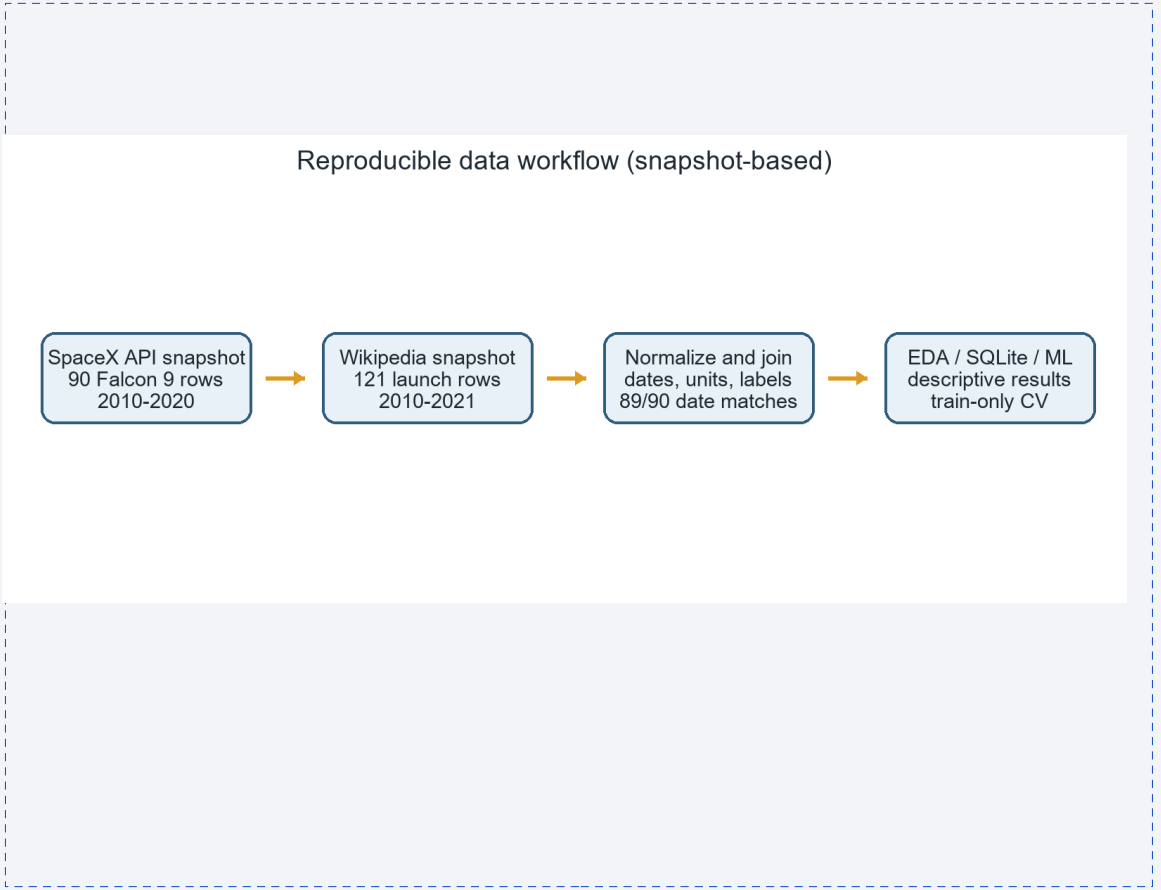
The SpaceX API endpoint did not respond during this rebuild. A fixed public course-project mirror was used instead.

Snapshots may differ from current API and Wikipedia records; the two tables are not interchangeable.

Gihub repository： https://github.com/luna029/public_popopopopopo

Data Sources and Reproducible Flow

- Source A: API-derived launch snapshot (90 rows).
- Source B: historical web-table snapshot (121 rows).
- Normalize date formats, mass units, and outcome labels.
- Join on launch date, not Flight_Number: 89/90 API rows matched.
- One API event (2016-09-01) is absent from the web table.



Data Collection - Scraping

API Dataset: Fields and Limits

- Analysis table: 90 launches, 2010-2020; target is successful first-stage landing.
- Success = True ASDS, True RTLS, or True Ocean. All other outcomes, including no attempt, are non-success.
- Inputs: Flight_Number, Payload_Mass, Flights, Reused_Count, Orbit, Launch_Site.
- Landing_Pad has 26 missing values; it is not used as a model feature.
- The API-derived snapshot drives EDA, mapping, dashboard, and model evaluation.

Place your flowchart of web scraping here

Data Wrangling

Parsed dates to ISO format and converted payload masses to kilograms.

Web table: 3 of 121 payload entries are missing or not parseable; SQL payload totals use numeric values only.

Standardized landing outcomes into success, failure, and no-attempt/other categories for SQL summaries.

Joined snapshots on date because Flight_Number values use different numbering schemes.

89 date matches; the unmatched 2016-09-01 API launch is retained only in API-based analysis.

Exploratory Data Analysis

Visualized launch sequence by site and landing result.

Compared payload mass across sites and orbit classes.

Calculated landing success rates by orbit and by year.

Checked patterns at group level; small groups and changing mission mix limit comparisons.

No causal claim is made from observational data.

SQL Analysis

Loaded the 121-row web-table snapshot into SQLite.

Queries use SELECT, DISTINCT, LIKE, SUM, AVG, MIN, WHERE, GROUP BY, and ORDER BY.

Payload text is parsed to numeric kilograms before aggregation.

Queries reproduce launch-site, customer, booster, landing, payload, and date summaries.

Web-table outcomes are not the same target definition as the 90-row classifier.

Folium Mapping

Mapped three API launch sites from supplied coordinates with Folium and OpenStreetMap tiles.

Added outcome markers and a Florida-area inset for the two Cape Canaveral launch complexes.

The inset line measures approximately 5.76 km between supplied site coordinates.

This is coordinate-to-coordinate distance; road access and coastline distance were not modeled.

Interactive Visual Analytics

Plotly Dash view over the 90-launch API-derived snapshot.

Controls: analysis view, launch site, and payload range.

Views: successful landings by site; outcome mix by selected site; payload vs. launch sequence.

No-attempt launches are grouped with non-success and labeled explicitly.

Slides 39-41 show the controls and representative filters.

Classification Methodology

Target: first-stage landing success (60 positive, 30 non-success cases).

Inputs: Flight_Number, Payload_Mass, Flights, Reused_Count, Orbit, Launch_Site; preprocessing is fit within each pipeline.

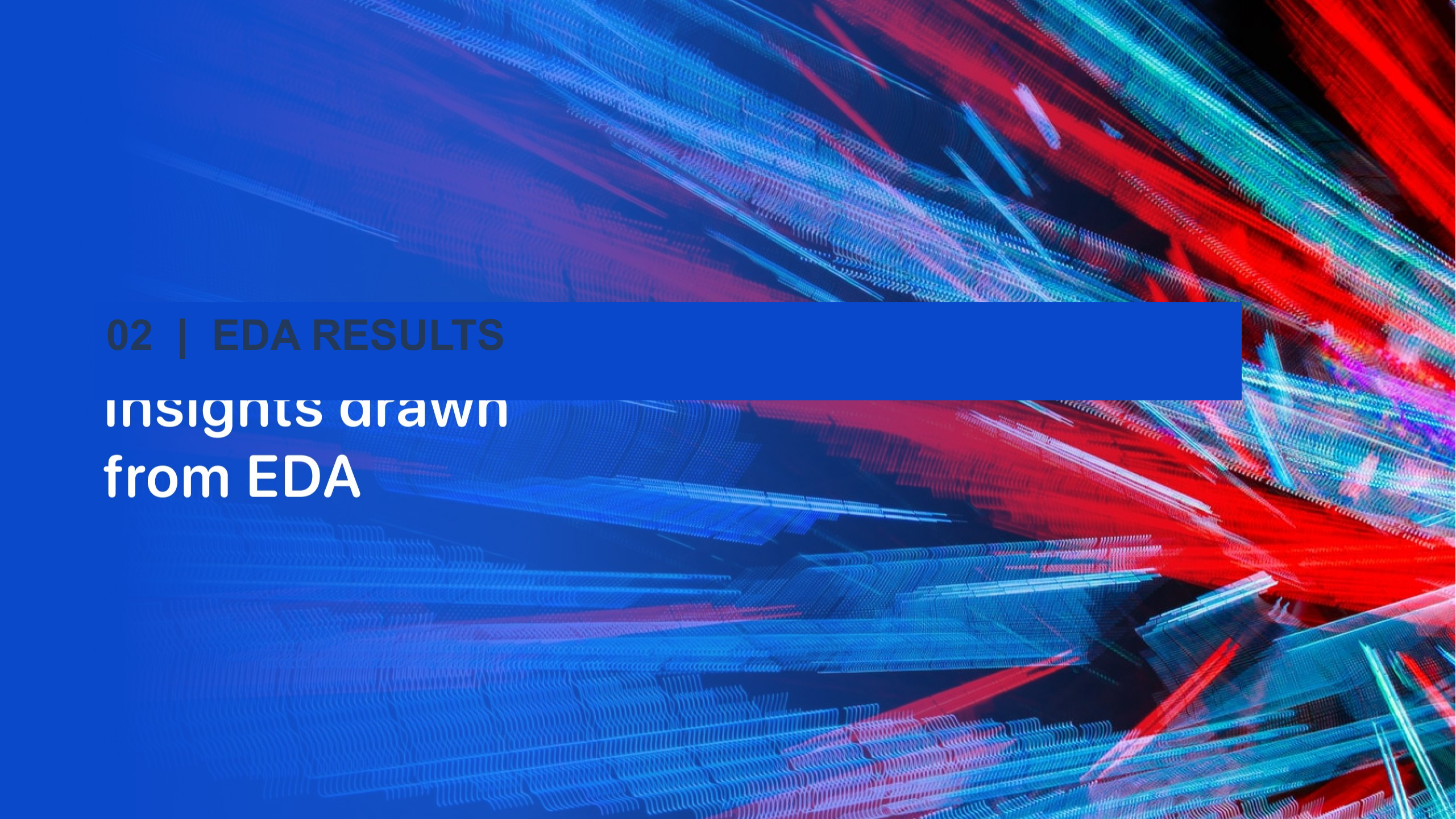
Stratified 80/20 split (random_state=42); model selection uses repeated stratified 5-fold CV x 10 on training data only.

Compare Logistic Regression, Decision Tree, KNN (k=5), SVM (RBF), and Random Forest.

Random holdout is not time-aware; future use requires temporal validation and more data.

Results at a Glance

- 60/90 API-snapshot launches landed successfully (66.7%).
- Site success rates range from 60.0% at SLC 40 to about 77% at KSC and VAFB.
- Orbit and yearly rates vary substantially; group sizes and mission mix differ.
- SQLite web-table queries reveal customer and booster payload patterns.
- Random Forest leads CV (76.2%); held-out accuracy is 88.9% on only 18 launches.

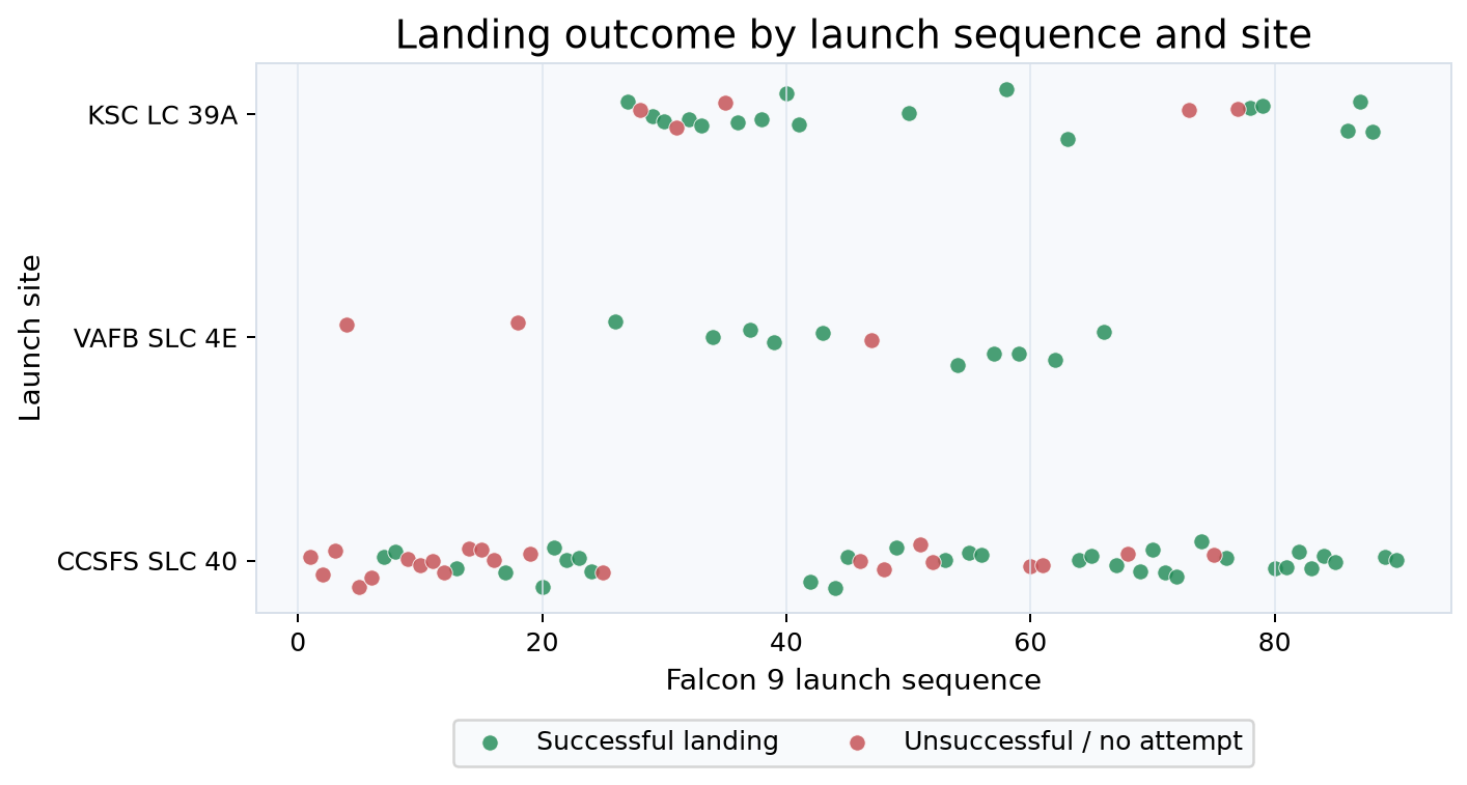


02 | EDA RESULTS

Insights drawn from EDA

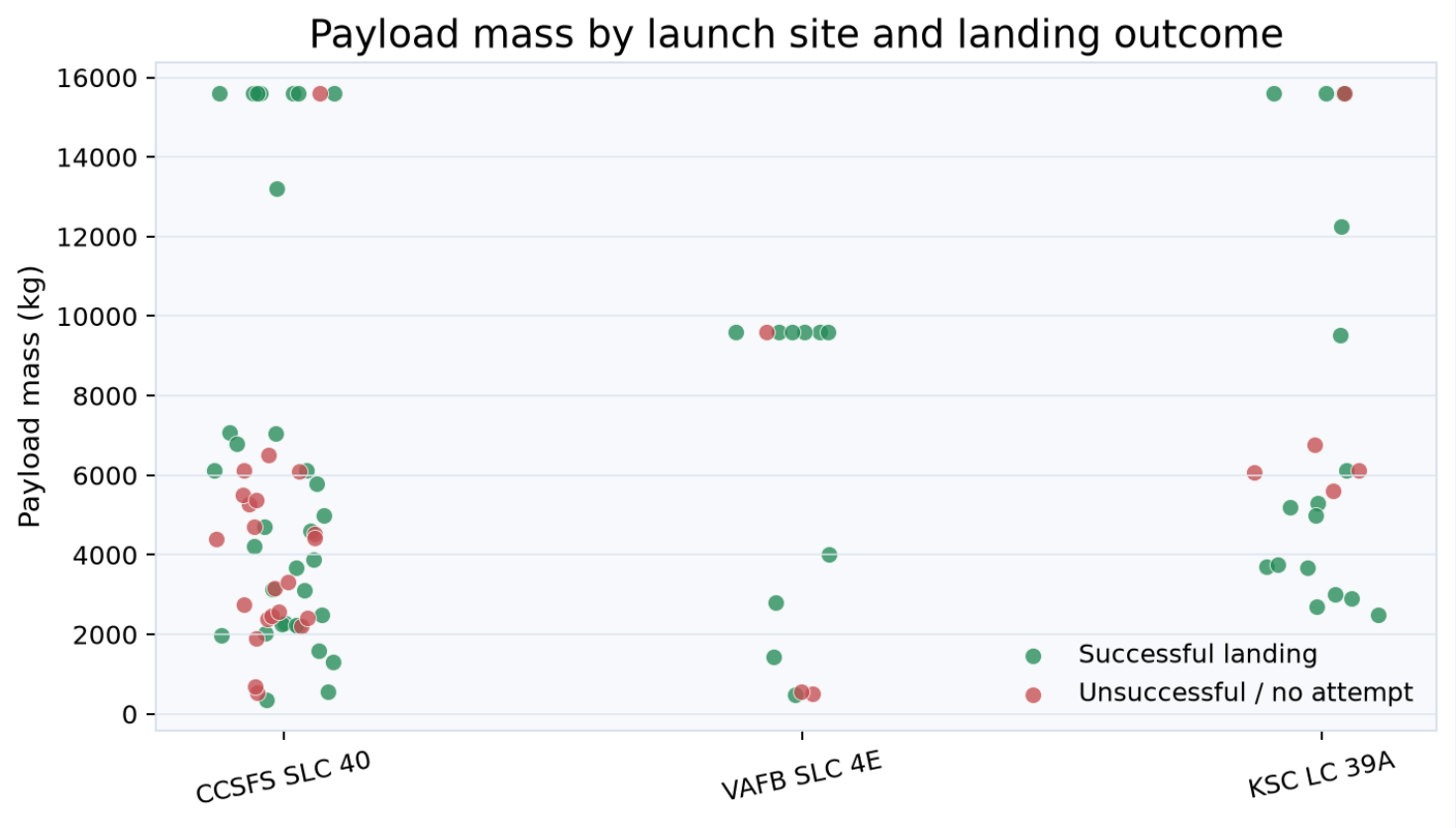
Landing Outcome by Launch Site and Sequence

- 90 launches across three sites.
- SLC 40 has the largest sample (55); KSC 22; VAFB 13.
- Later sequences show more successful recoveries, consistent with an evolving program.
- Sequence tracks time and mission mix; this is not a causal effect.



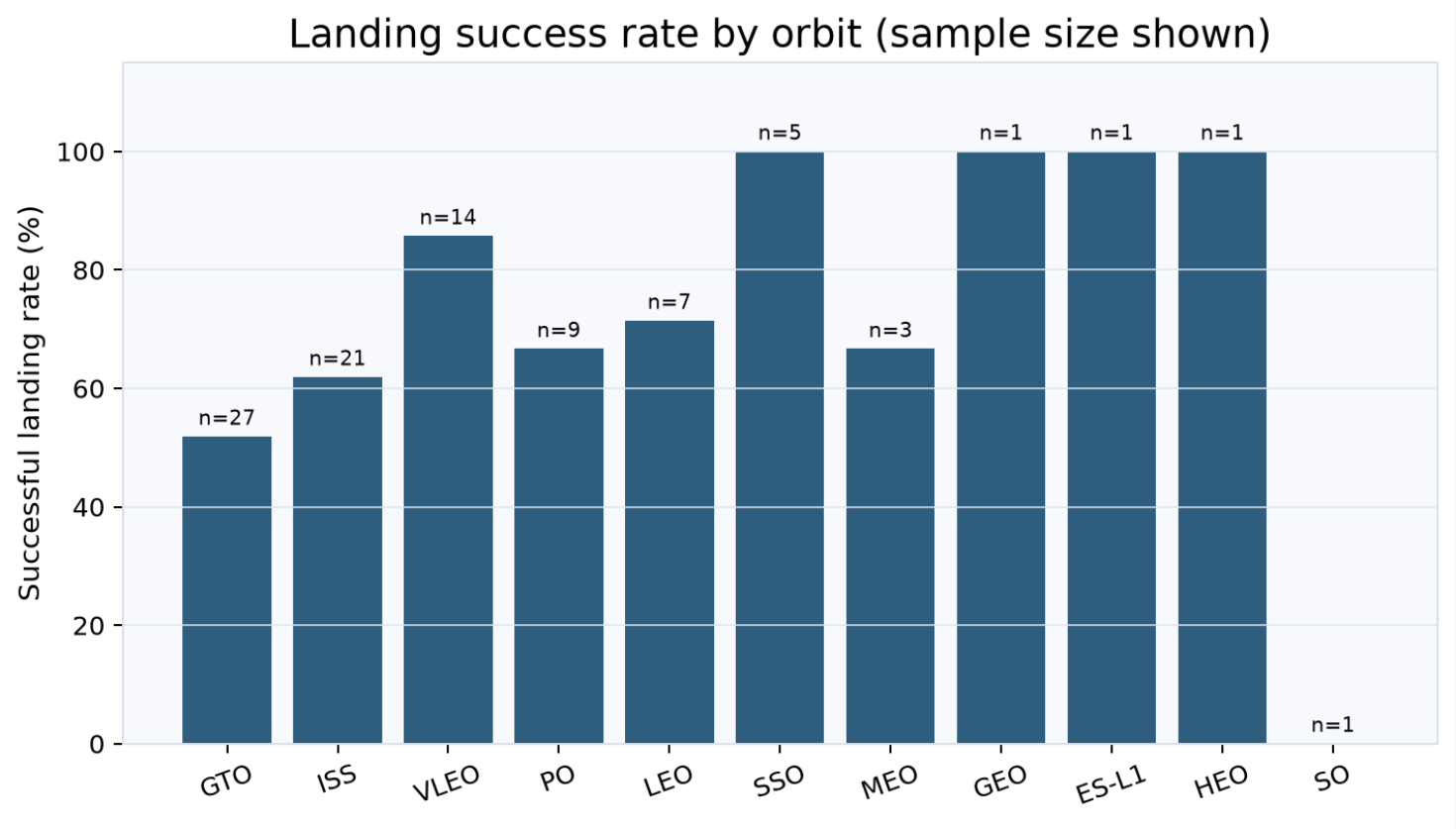
Payload Mass by Launch Site

- Payload distributions overlap across sites.
- Site-level differences reflect mission assignment as well as operations.
- Payload mass alone does not determine recovery outcome.
- Mass values are in kilograms.



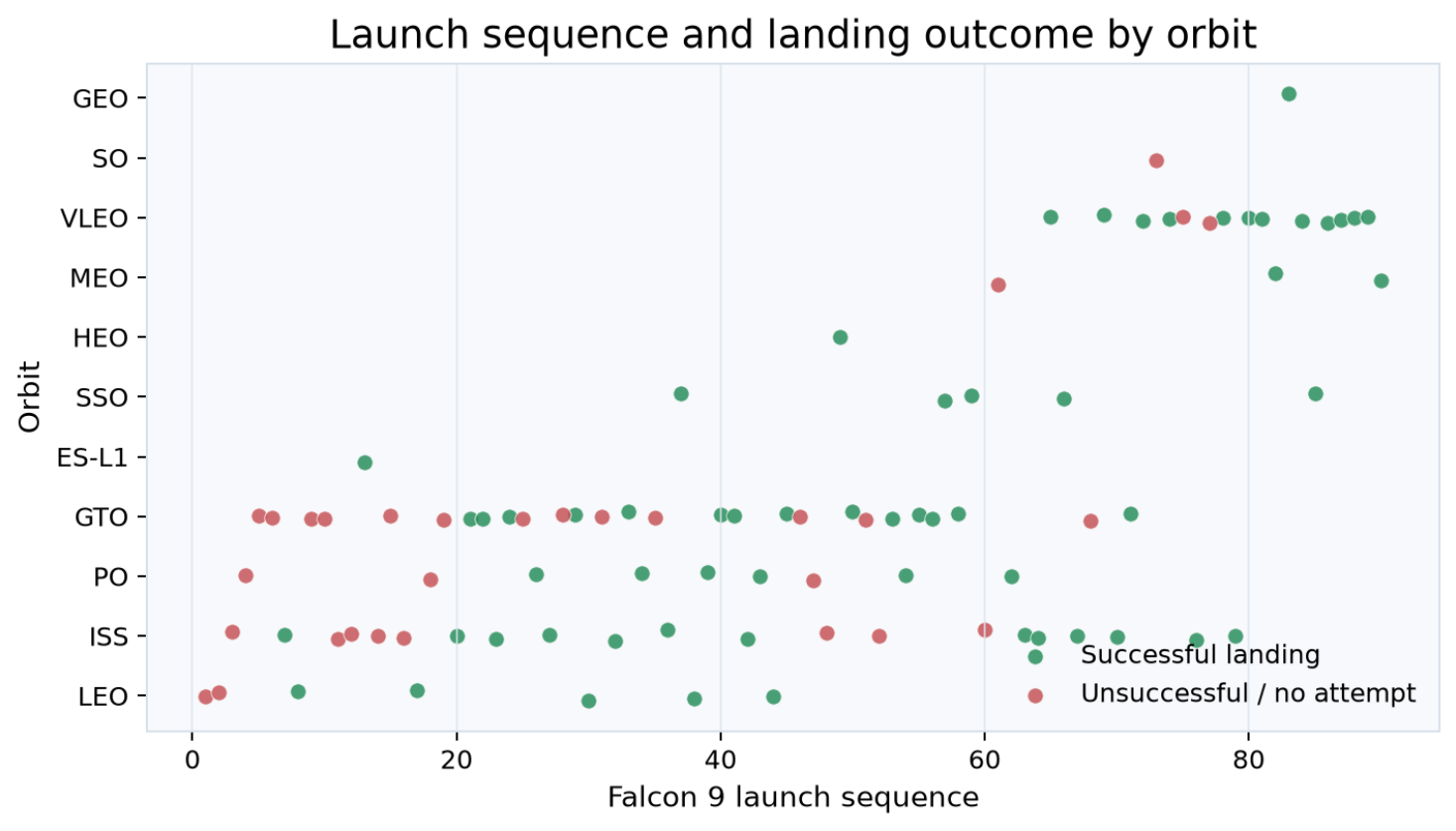
Landing Success by Orbit

- SSO: 5/5 successful; VLEO: 12/14 (85.7%).
- GTO: 14/27 (51.9%).
- Rates are descriptive and orbit groups are uneven in size.
- Launch era and mission type may confound these comparisons.



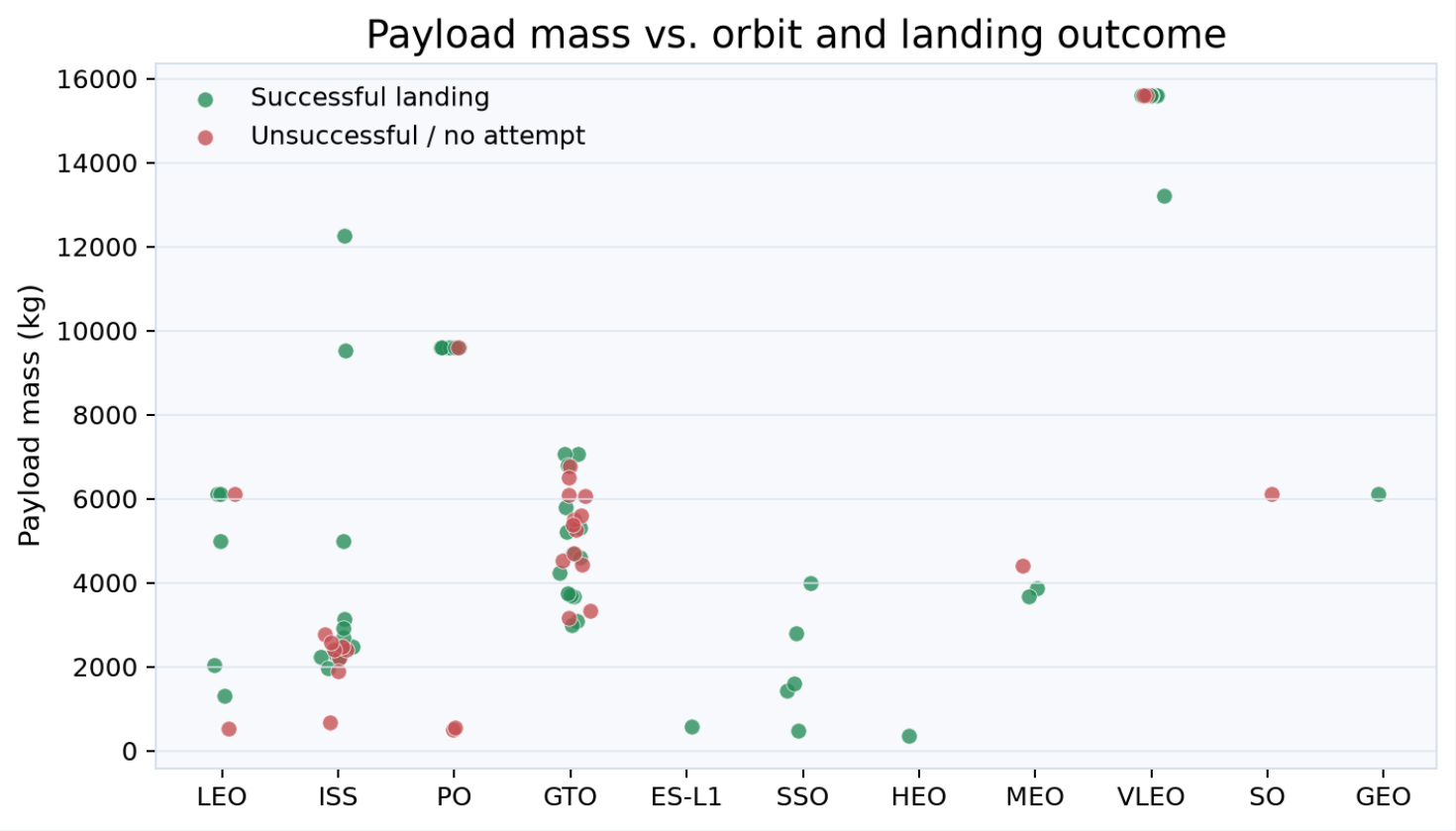
Flight Sequence by Orbit

- Orbit mix changes across the launch sequence.
- Later launches include more recurring LEO/Starlink missions.
- Sequence is a time proxy, not an independent causal factor.



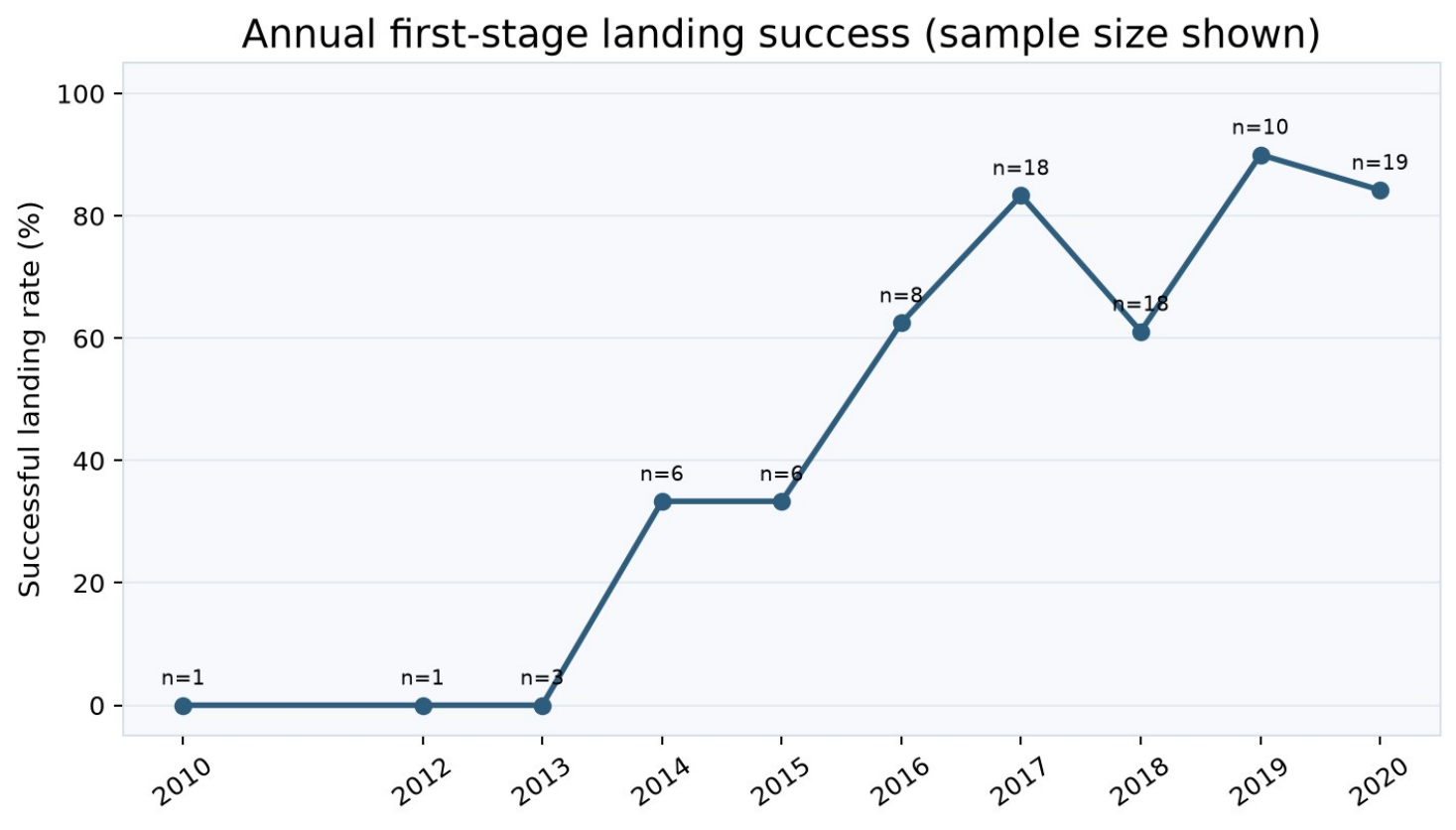
Payload Mass and Orbit Outcomes

- Payload mass varies widely within orbit categories.
- The 15,600 kg payload is the maximum in the web-table snapshot (24 tied rows).
- Different orbit missions have different payload profiles; avoid a single cutoff.



Annual Landing Success Trend

- Early attempts were mostly unsuccessful or not attempted.
- Annual success rates rise in later years, reaching roughly 80-90% in several recent years.
- Yearly sample sizes are small; this is descriptive, not a forecast.



Distinct Launch-Site Labels

Five labels: CCAFS, CCSFS, Cape Canaveral, KSC, VAFB.

SQL

```
SELECT DISTINCT Launch_Site FROM spacex_web_scraped;
```

Some names refer to the Cape Canaveral area.

First Five CCAFS Records

Flights 1-5, dated 2010-06-04 through 2013-03-01.

SQL

```
SELECT Flight_Number, Launch_Site, Date_iso
FROM spacex_web_scraped WHERE Launch_Site LIKE 'CCA%'
ORDER BY Flight_Number LIMIT 5;
```

There are 25 CCAFS-labeled launches in the table.

NASA Customer Payload Total

124,708 kg across 32 rows.

SQL

```
SELECT SUM(Payload_kg)
FROM spacex_web_scraped WHERE Customer LIKE '%NASA%';
```

Aggregate uses numeric payload masses.

Average Payload for F9 v1.1

2,534.7 kg (n=15).

SQL

```
SELECT COUNT(*), AVG(Payload_kg)
FROM spacex_web_scraped WHERE Booster_Version LIKE '%F9 v1.1%';
```

Mean is calculated in kilograms.

First Successful Ground Landing

2015-12-22.

SQL

```
SELECT MIN(Date_iso)
FROM spacex_web_scraped WHERE Landing_Outcome = 'Success (ground pad)';
```

First successful ground landing recorded by this table.

Drone-Ship Successes with 4,000-<6,000 kg Payload

10 flights: 24, 28, 32, 43, 60, 63, 64, 88, 89, 97.

SQL

```
SELECT Flight_Number, Payload_kg, Date_iso
FROM spacex_web_scraped
WHERE Landing_Outcome='Success (drone ship)'
AND Payload_kg >= 4000 AND Payload_kg < 6000;
```

Interval includes 4,000 kg and excludes 6,000 kg.

Landing Outcome Counts

85 success; 12 failure; 24 no-attempt/other (121 rows).

SQL

```
SELECT CASE
  WHEN Landing_Outcome LIKE 'Success%' THEN 'Success'
  WHEN Landing_Outcome LIKE 'Failure%' THEN 'Failure'
  ELSE 'No attempt/other' END AS outcome, COUNT(*)
FROM spacex_web_scraped GROUP BY outcome;
```

The web-table labels are combined into three groups; they are not identical to the model target.

Maximum Payload Rows

15,600 kg, tied across 24 rows; F9 B5 family.

SQL

```
SELECT Booster_Version, Serial, Payload_Mass_kg, Flight_Number
FROM spacex_web_scraped
WHERE Payload_Mass_kg = (
  SELECT MAX(Payload_Mass_kg) FROM spacex_web_scraped
);
```

Returns all rows tied at the maximum payload mass.

Failed Drone-Ship Landings in 2015

Flights 14 (Jan 10) and 17 (Apr 14).

SQL

```
SELECT Flight_Number, Booster_Version, Date_iso
FROM spacex_web_scraped
WHERE Landing_Outcome='Failure (drone ship)' AND Date_iso LIKE '2015-%';
```

Two records are returned for 2015.

Landing Outcomes: 2010-2017

Leaders: no attempt 11; success drone ship 5; failure drone ship 4; success ground pad 3.

SQL

```
SELECT Landing_Outcome, COUNT(*) AS n
FROM spacex_web_scraped
WHERE Date_iso BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY n DESC;
```

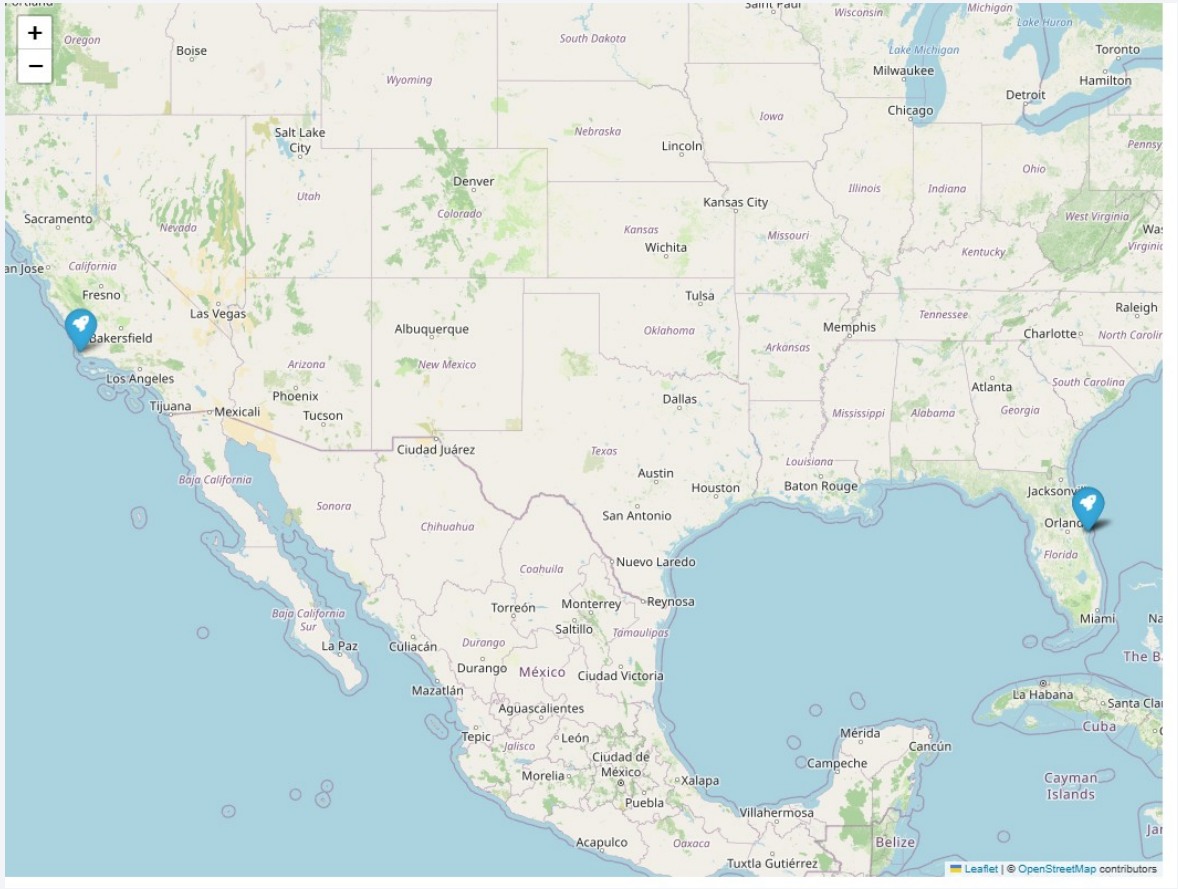
Several other labels occur at lower counts; wording retained from source.



03 | FOLIUM MAPS

Launch Sites Proximities Analysis

Launch Sites Across the United States

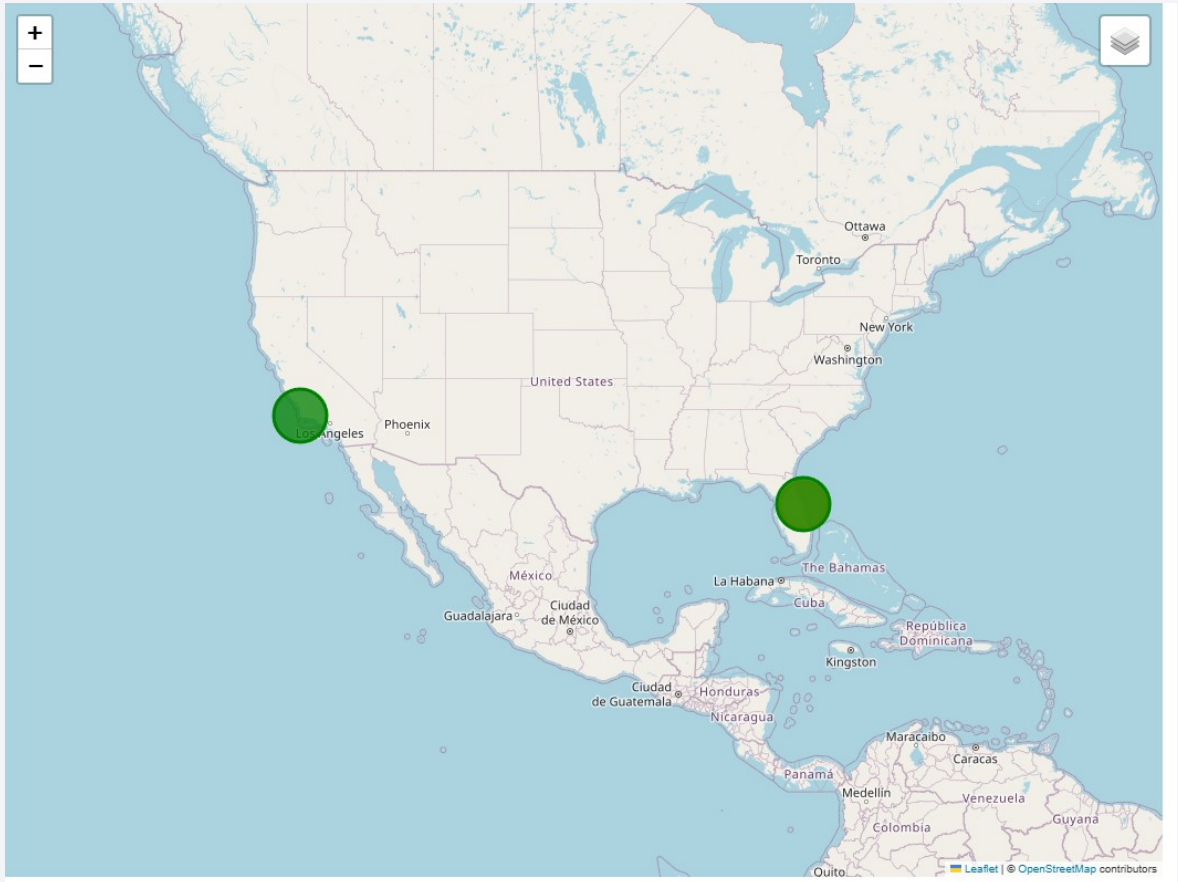


Three Falcon 9 launch sites are mapped from API-derived coordinates.

Cape Canaveral sites are in Florida; VAFB is in California.

OpenStreetMap tiles provide context.

Landing Outcomes by Site

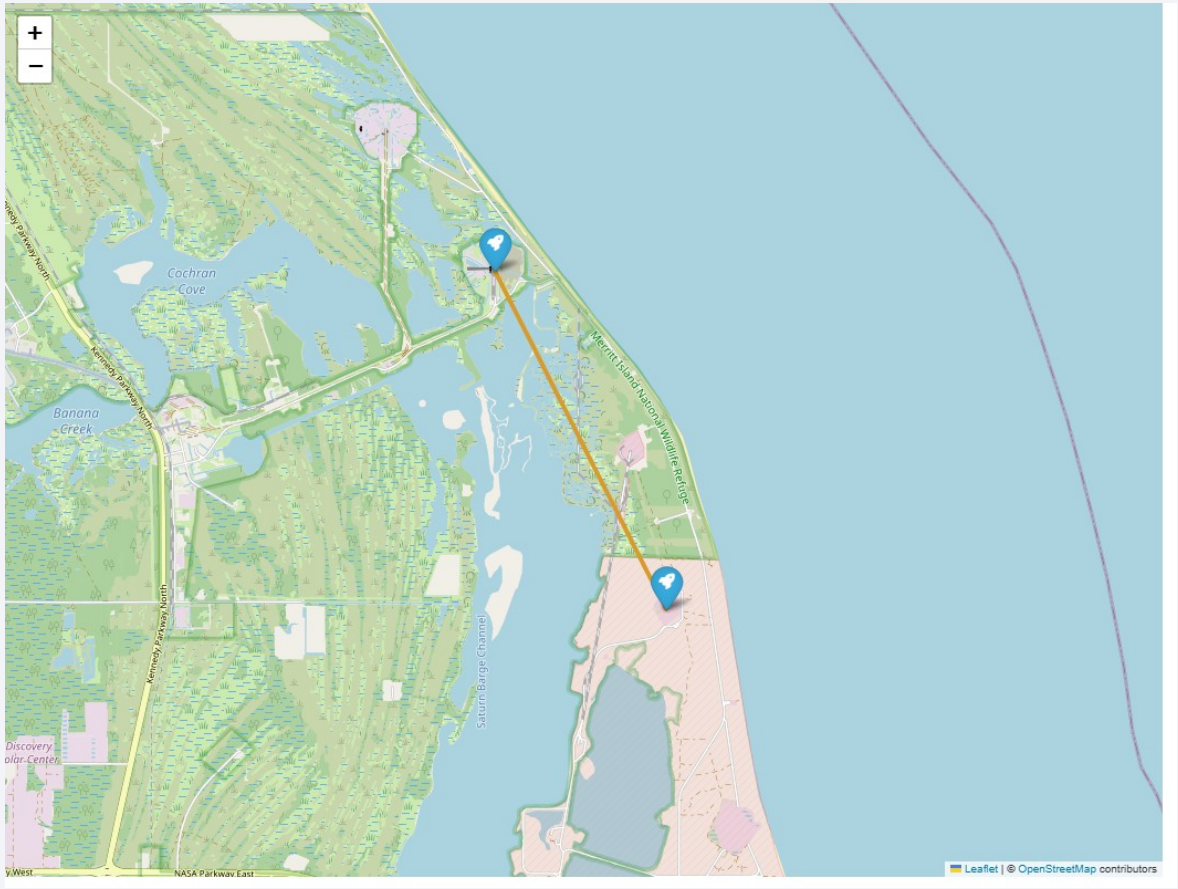


Markers show launch-site locations and aggregated outcomes.

Rates: SLC 40 60.0%; KSC 77.3%; VAFB 76.9%.

Differences are descriptive; counts and mission mix vary.

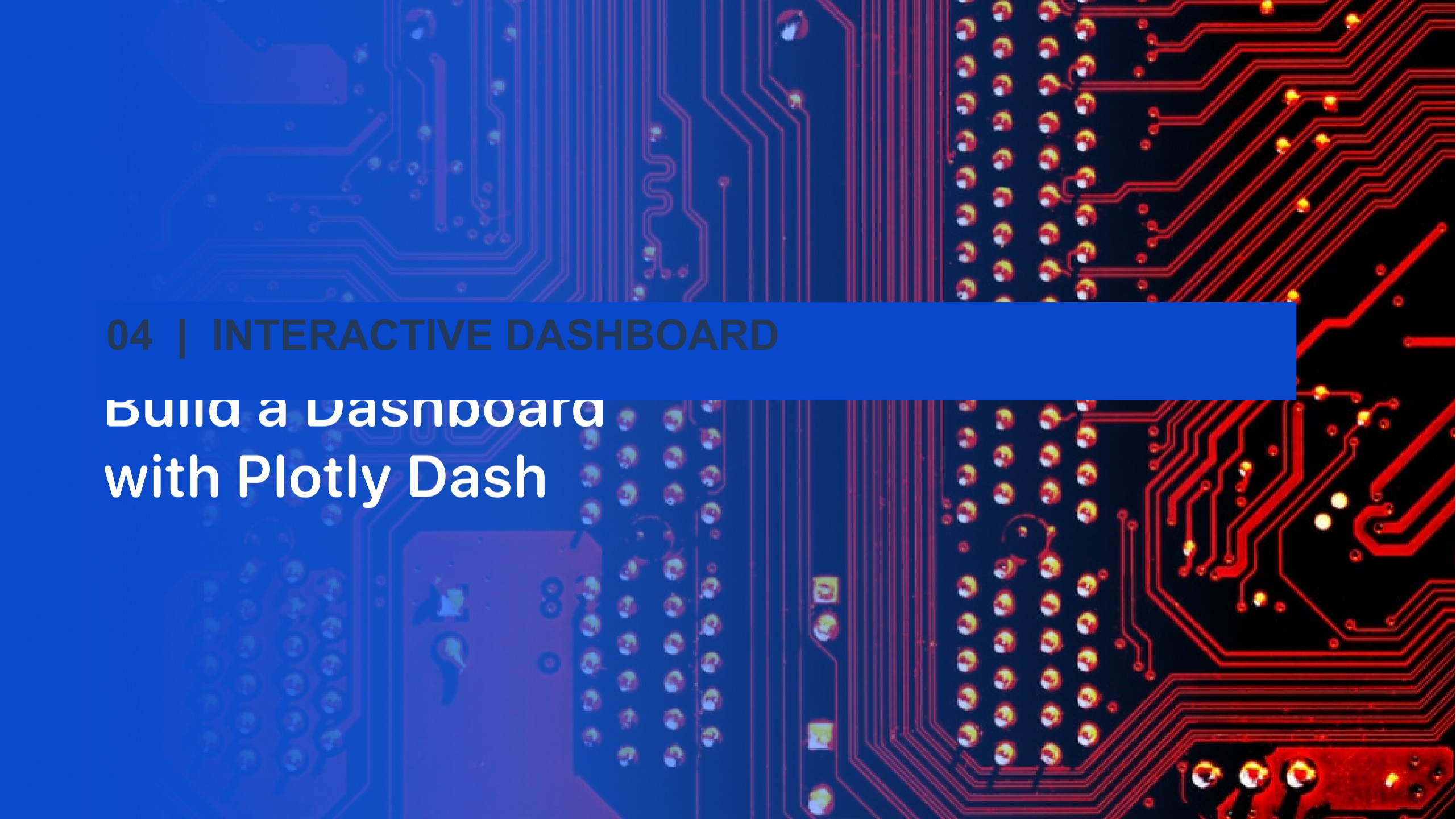
Cape Canaveral Launch-Site Detail



The inset distinguishes the two Florida launch complexes.

Straight-line distance between supplied coordinates: about 5.76 km.

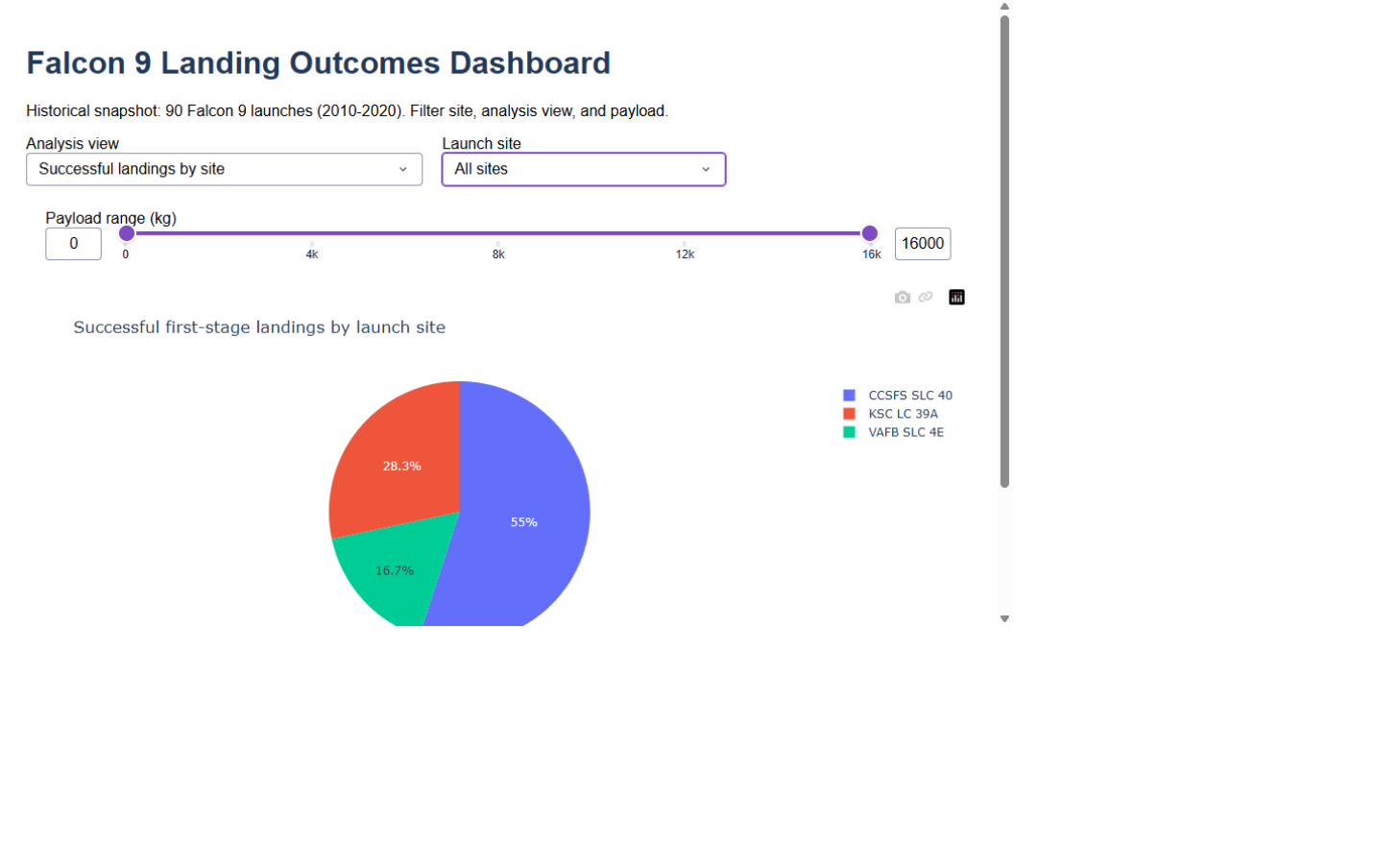
Not a road, coastline, or operational-access distance.



04 | INTERACTIVE DASHBOARD

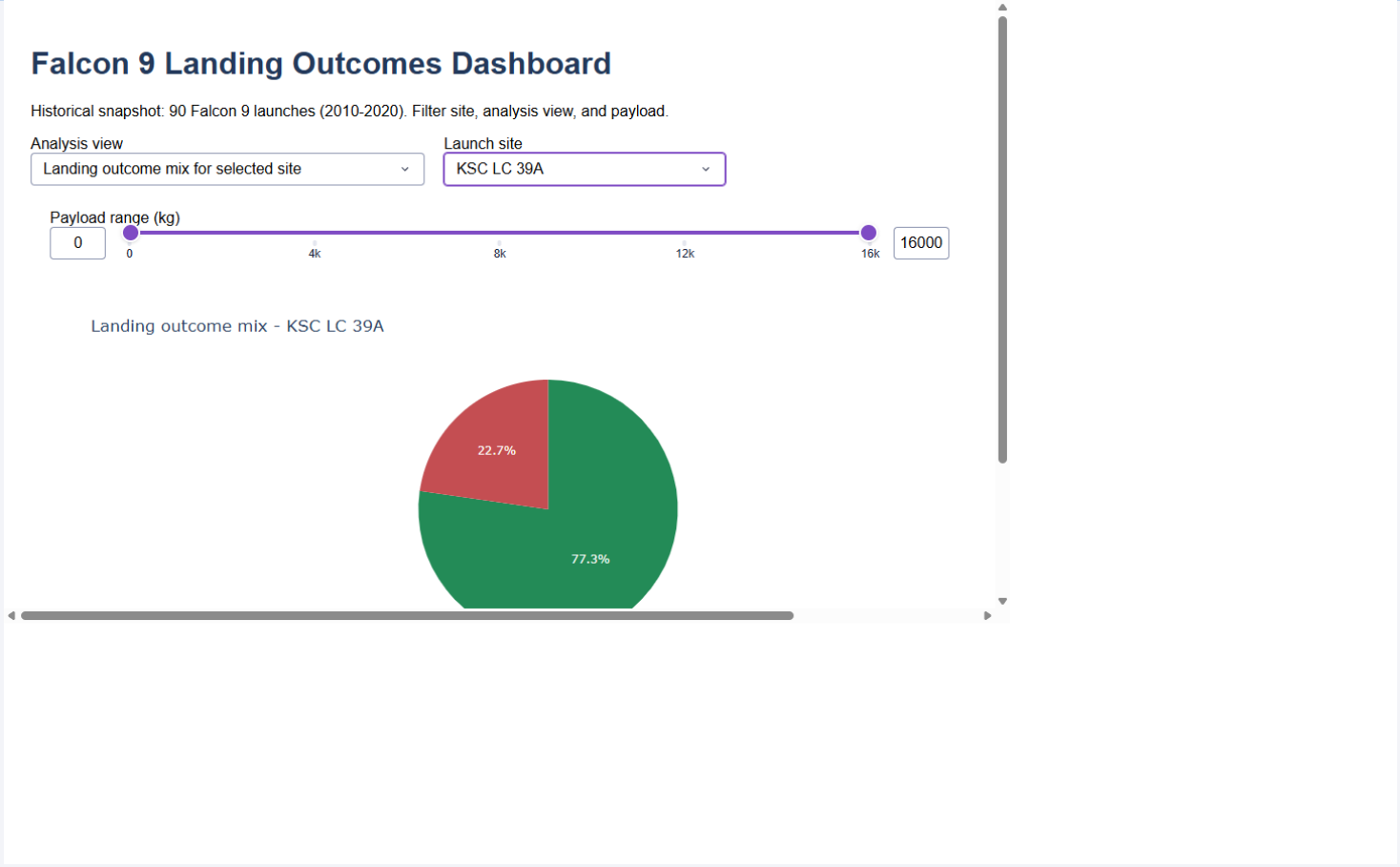
Build a Dashboard with Plotly Dash

Dash: Successful Landings by Site



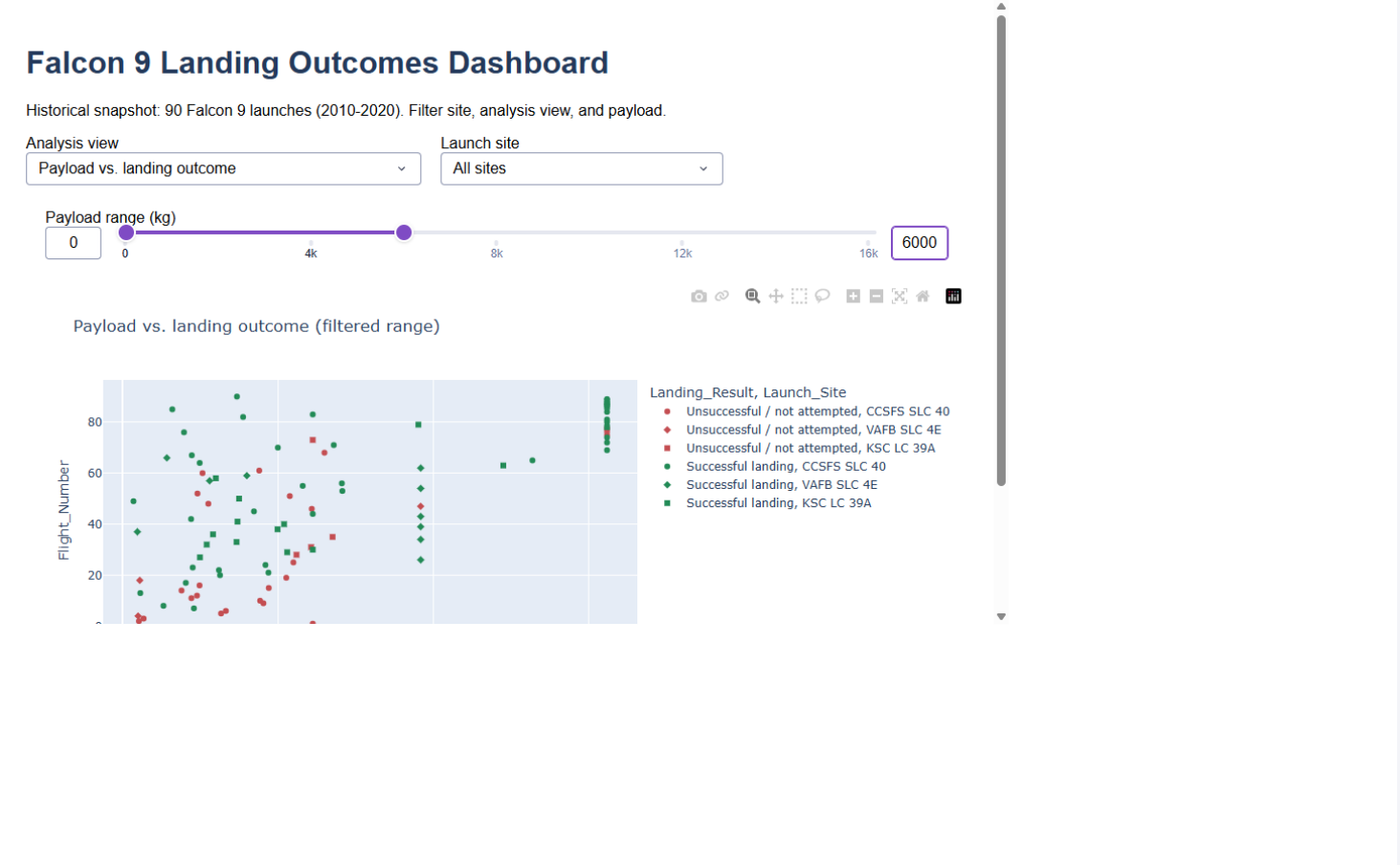
Successful landing counts: 33 at SLC 40, 17 at KSC, 10 at VAFB (60 total).

Dash: KSC Landing Outcome Mix



KSC: 17/22 successful (77.3%); 5/22 unsuccessful or not attempted. Descriptive, not causal.

Dash: Payload vs. Landing Outcome



Interactive scatter filtered to all sites and payloads up to 6,000 kg; color is outcome, symbol is site.

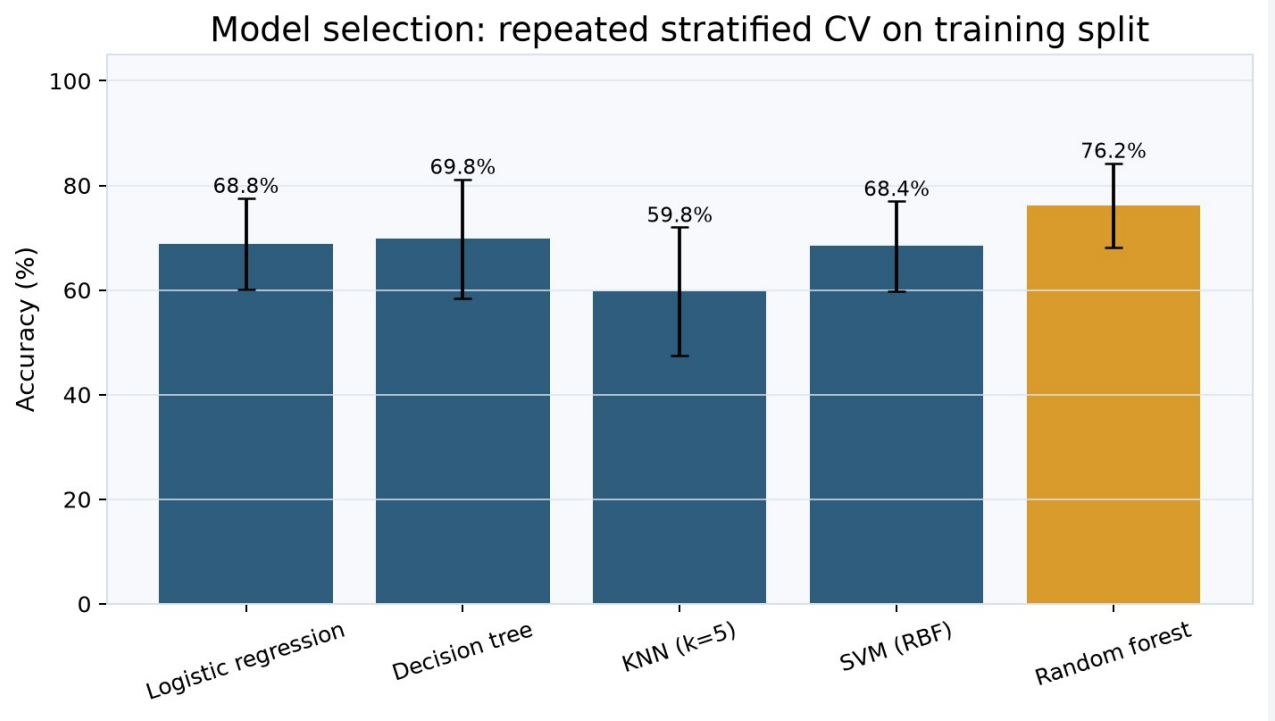


05 | CLASSIFICATION

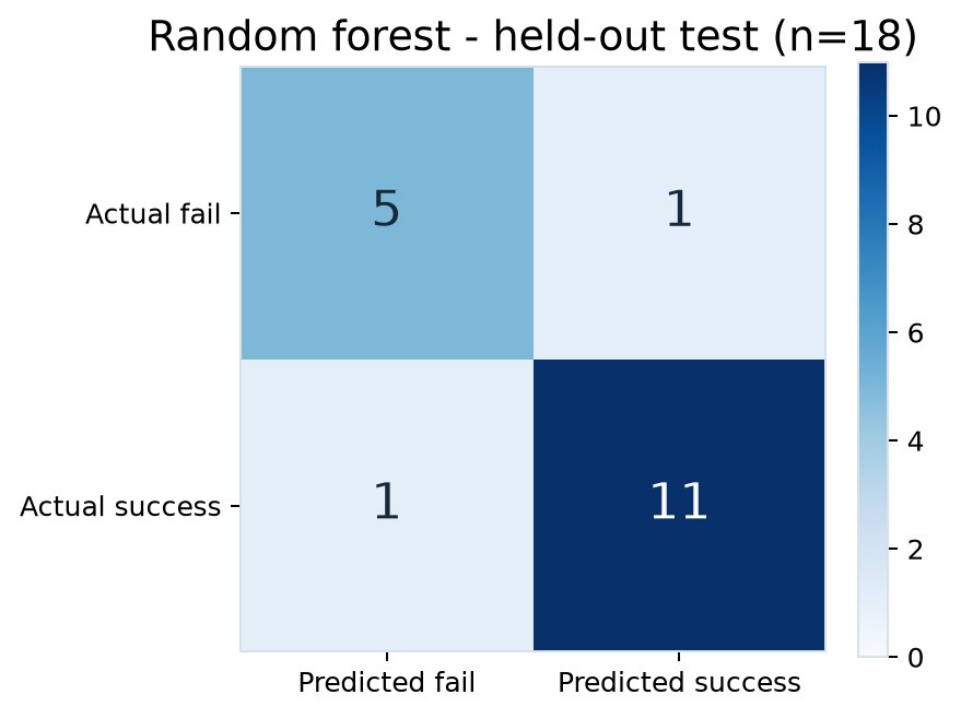
Predictive Analysis (Classification)

Model Comparison

- Repeated stratified CV mean accuracy on training split:
- Logistic Regression 68.8%; Decision Tree 69.8%; KNN 59.8%; SVM 68.4%; Random Forest 76.2% +/- 8.1%.
- Random Forest selected by CV; held-out accuracy 88.9%.
- The 18-case test set is small; CV and test scores answer different questions.



Random Forest Confusion Matrix



- Held-out test set: 18 launches.
- Actual fail predicted fail: 5; actual fail predicted success: 1.
- Actual success predicted fail: 1; actual success predicted success: 11.
- 16/18 correct (88.9%); one error in each class.
- Small test sample; not deployment performance.

Conclusions and Next Steps

A 66.7% landing success rate is observed across the 90-launch API-derived snapshot.

Key insight: SLC 40 contributes 55% of successful landings because it has the most launches, but its within-site success rate (60.0%) is lower than KSC and VAFB (about 77%).

Recovery outcomes improve across the historical sequence, but time and mission mix are confounded.

Random Forest leads training-only repeated CV (76.2%); its 88.9% test score is based on 18 cases.

Next: obtain a current documented dataset; use time-ordered validation; report class-wise metrics; test external data.

Appendix: Sources, Definitions, and Limitations

API-derived CSV mirror (fixed commit): https://github.com/jgp-13/SpaceX_project/tree/8b16eb3e32af9b46fd6d279c7b344b3db6bb2631

SpaceX API project: https://github.com/r-spacex/SpaceX-API

Web-table source: https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

API success: True ASDS / True RTLS / True Ocean; all other categories are non-success.

Model: 90 cases; stratified 80/20 split (seed 42); repeated stratified 5-fold x 10 CV on training rows; pipeline preprocessing.

Caveats: snapshots may be stale; web-table labels are heterogeneous; Flight_Number values do not align; 89/90 date matches; test n=18; random split is not temporal.

Reproducible public-data draft. No user-provided notebook, GitHub submission URL, or presenter name was supplied; none is invented.

Gihub repository： https://github.com/luna029/public_popopopopopo

Thank you!

